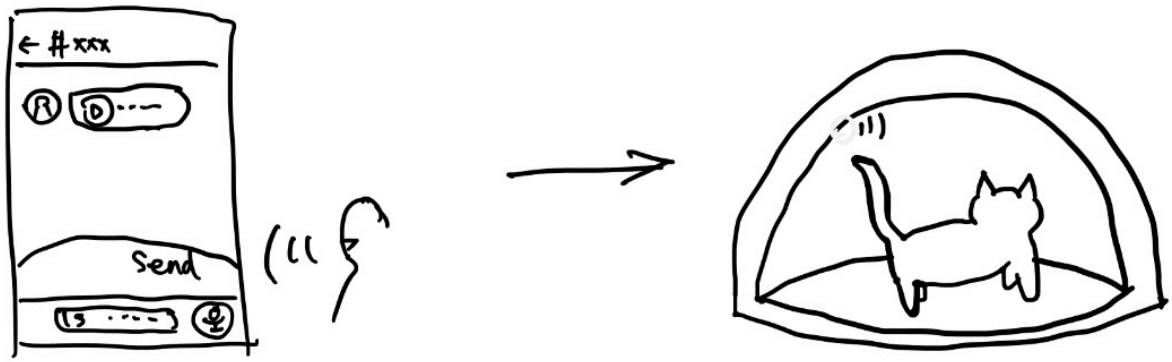


Not Alone, Just Apart: Designing PurrPost to Mitigate Pet-Owner Separation Anxiety

Duration	01/2025 – 06/2025
Methods	Autoethnography, Research through Design, Sensor-Based Interaction, Survey
Themes	Animal-Computer Interaction (ACI), Human-Animal Interaction, Hybrid Collaboration
Type	Research Project, Thesis

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Full Text: Text

Code: GitHub Repository

Overview

Physical separation can cause distress for pets and their owners, yet technologies supporting their emotional connection remain limited.

This study investigates how technology can help mitigate separation anxiety in remote pet-owner relationships. Using research through design, PurrPost, a two-way communication system connecting a cat and her remote owner, is developed, and its use is examined through a one-month autoethnography involving daily diaries, self-assessments, and behavioural observations.

The findings highlight four key aspects: temporal patterns of separation anxiety, technical reliability, mutual engagement and privacy, and the roles of co-located family members and animal welfare.

The study contributes four design guidelines for remote human-animal interaction: supporting different phases of separation, ensuring technical robustness, accommodating delay and ambiguity, and protecting privacy during emotional expression.

https://notion-analytics.com/t?doc=purrpost

Why this matters

- Physical separation can cause anxiety for both pets and their owners, yet interventions addressing both sides remain limited.
- Existing remote pet technologies largely focus on one-way monitoring, with less attention to reciprocal interaction and emotional connection.
- Research on hybrid interaction has primarily focused on humans, leaving the roles of animals and co-located caregivers underexplored.

💡 RQ: **How can technologies help mitigate separation anxiety in remote pet-owner relationships for both pets and their owners?**

Design

A **Research through Design (RtD)** approach was adopted to investigate technology-mediated pet-owner interaction through iterative prototyping and reflection.

PurrPost was designed to support a two-way emotional connection between a cat and her remote owner.

Design requirements

1. **Safety and comfort:** Prioritise the cat's welfare without disrupting her natural behaviour or daily routine.
2. **Owner reassurance:** Reduce the owner's concerns about the cat's well-being during separation.
3. **Pet reassurance:** Provide a safe, comfortable environment or enrichment to help redirect the cat's attention from separation.

Design rationale

The design translated these requirements into three main components:

1. **A familiar safe space:** Integrate the prototype into the cat's existing nest, avoiding unfamiliar environments and intrusive wearable devices.
2. **Status detection and notifications:** Use force sensors to detect the cat's presence and activity in the nest, automatically sending updates through a Discord bot from the cat's perspective.
3. **Two-way audio interaction:** Allow the owner to send voice messages and receive recordings of the cat's natural sounds, using familiar voices and gentle audio rather than potentially stressful tactile or visual stimulation.

Daily activity summaries and weather updates provided additional context about the cat's routine.

Prototype iteration

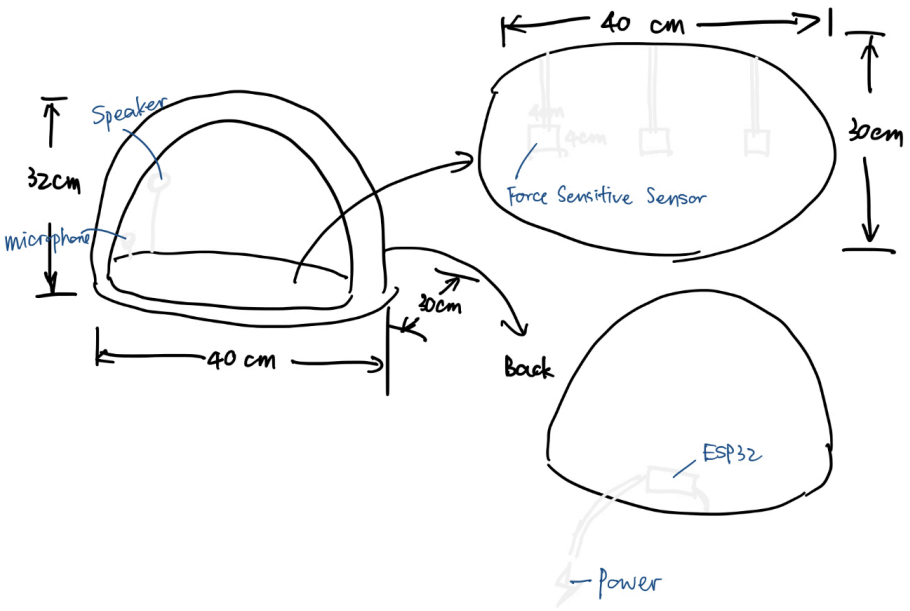


Figure 5.1: Sketch of the initial PurrPost design concept.

Initial testing revealed incomplete sensor coverage and exposed wires that attracted the cat's attention. The design was revised by positioning the force sensors between cardboard layers and embedding the microphone and speaker beneath the lower layer to improve detection and robustness.

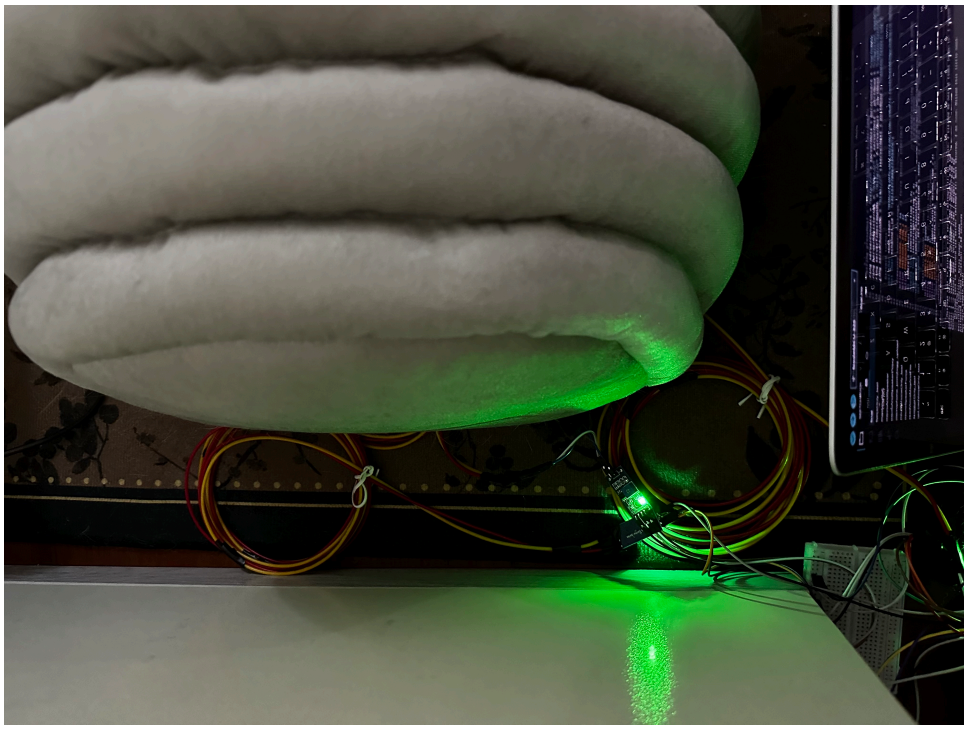


Figure 5.2: Wires placed behind the cat nest in the initial implementation.



Figure 5.3: Components embedded beneath cardboard in the revised implementation.

Study

Research approach

A one-month **autoethnographic study** was conducted to examine emotional and behavioural changes in a remote pet-owner relationship during physical separation.

Study setting & timeline

March 26 – May 1, 2025

The study involved one owner-cat relationship and two family caregivers.

- **Before separation (March 26–29):** The owner and cat were together in Shanghai, China.
- **Separation (March 30):** The owner relocated to Copenhagen, Denmark, while the cat remained in Shanghai with two family caregivers.
- **Unscheduled interaction (March 26 – April 7):** The owner sent audio messages at varying times.
- **Scheduled interaction (April 8 – May 1):** Audio interaction followed a daily schedule at 15:30 Copenhagen time.

During remote interactions, the caregivers observed the cat's responses and shared feedback, photos, and videos through WeChat. The owner also monitored the cat through a camera.

Data collection

- **Pre-study interviews:** Brief interviews with co-located family caregivers to understand the cat's habits and behavioural changes (March 15–25).
- **Daily diary:** The owner's emotions, reflections, and experiences of separation and interaction.
- **Photos and videos:** Observations of the cat's behaviour, including caregiver reports and camera observations.
- **Discord messages:** Text and audio interactions between the owner and the system.
- **Five self-assessments:** Adapted separation-anxiety questionnaires completed on March 19, March 29, April 7, April 15, and April 23.

- **Interaction statistics:** Daily audio-message counts and the frequency and duration of the cat's visits to the nest or cushion.

Data analysis

- **Thematic analysis** of diary entries, contextualised by photos and videos, to identify emotional and behavioural patterns.
- **Descriptive analysis** of interaction statistics and questionnaire scores to examine changes over time.

Findings

1. Separation anxiety changed over time

Both the owner and the cat showed signs of separation-related anxiety, most noticeably before and shortly after separation, followed by a gradual decline.

- **Owner:** Self-assessment scores peaked before separation and declined to 0.1 approximately three weeks afterward. Audio-message activity also generally decreased over time.

Time	Mean score
Before prototype use	0.5
After prototype implementation, before separation	**1.1**
Approximately one week after separation	0.3
Approximately two weeks after separation	0.2
Approximately three weeks after separation	**0.1**

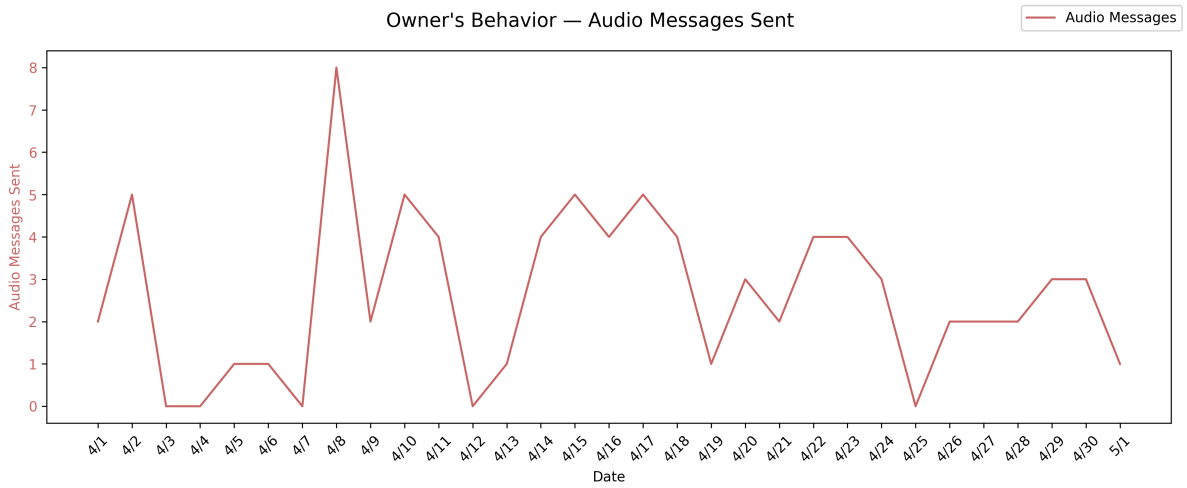


Figure 6.1: Number of audio messages sent by the owner.

- **Cat:** Unusual behaviours, including increased vocalisation, door-pawing, and restlessness, were observed around separation. Visits to the nest or cushion generally declined over time.

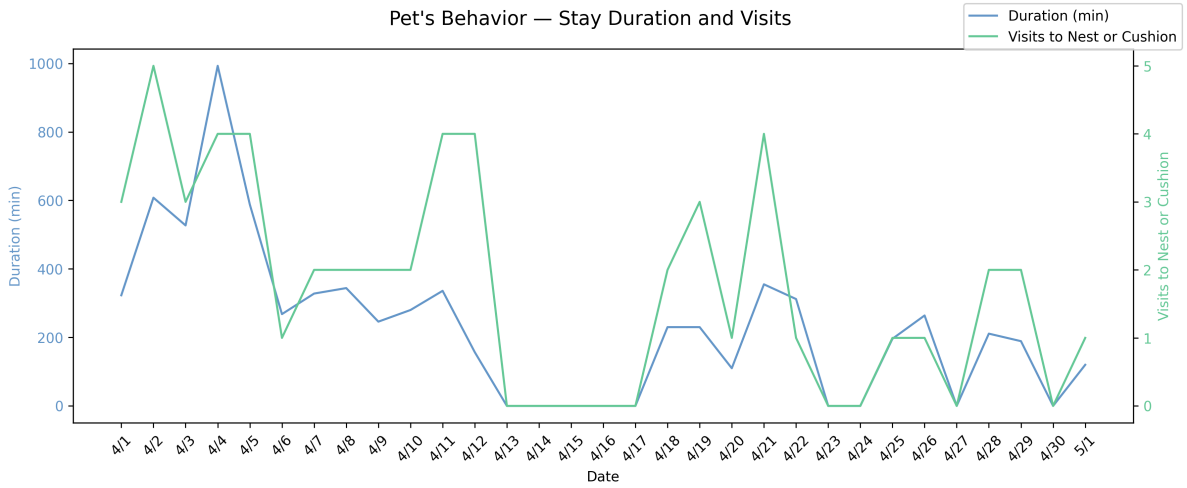


Figure 6.2: The cat's stay duration and number of visits to the nest or cushion.

2. Message presence mattered more than content

The timing and frequency of messages mattered more to the owner than their exact wording. Receiving messages helped maintain a sense of emotional connection.

Technical interruptions could disrupt this connection, making reliability particularly important. Delayed messages also influenced the owner's emotional engagement.

3. Mutual engagement and privacy shaped interaction

The owner's willingness to interact depended on the cat's perceived responsiveness, while the owner's interaction could also influence the cat's behaviour.

Privacy also mattered. The owner felt more comfortable expressing emotions when physically alone, rather than in the presence of others.

4. Co-located family members supported connection

Family caregivers helped operate the system, observe the cat, and coordinate remote interactions. Their involvement also strengthened family relationships and connections with the cat. However, differences in interpreting the cat's behaviour revealed challenges in interspecies communication.

5. Animal welfare required careful consideration

Remote audio interaction raised concerns about the cat's comfort and well-being. The cat's responses to unfamiliar sounds or the owner's disembodied voice were not always clear, highlighting the need to avoid potentially stressful interactions and respect the animal's natural behaviour.

Design Guidelines

1. Design for different phases of separation

Consider how emotional needs change before, during, and after physical separation. Support should begin before separation occurs, rather than focusing only on the period afterward.

2. Ensure technical robustness

Reliable message delivery and audio playback are essential for maintaining emotional connection. Technical failures can disrupt not only system functionality but also the owner's sense of connection with the pet.

3. Accommodate delay and ambiguity

Delayed responses and ambiguous feedback should not always be treated as problems. They can support slower, more reflective interactions that fit naturally

into daily routines without requiring continuous attention.

4. Support privacy in emotional expression

Privacy influences the owner's willingness to engage and express emotions. Design should accommodate both shared and private interactions, allowing owners to communicate comfortably in different social contexts.

Open Design Challenge: Interspecies Communication

Understanding animal behavior remains difficult without a shared language. Remote interaction often relies on co-located caregivers to observe, interpret, and coordinate communication, introducing potential misunderstandings and subjective bias.

Further research is needed before concrete design guidelines for interspecies communication and mediation can be established.

Limitations & Future Work

Limitations

- **Single-case study:** The study examined one owner-cat relationship in a specific hybrid setting, limiting the generalisability of the findings.
- **Assessment of separation anxiety:** The adapted owner questionnaire was not intended for clinical diagnosis, while the cat's emotional state was inferred primarily from behavioural observations without standardised indicators.
- **Caregiver perspectives:** Although co-located family members played essential roles, their experiences were not formally collected through interviews or questionnaires during the empirical study.
- **Interspecies understanding:** Interpreting animal behaviour remained dependent on human observations and assumptions, introducing potential bias and uncertainty.

Future Work

- **Broader studies:** Include more owners, caregivers, households, and animal species across different settings.

- **Improved assessment:** Develop more context-appropriate measures of owner anxiety and more systematic methods for evaluating animal behaviour and welfare.
- **Caregiver involvement:** Formally investigate caregivers' experiences through interviews or questionnaires.
- **Participatory design:** Explore co-design with pet owners, caregivers, and animal experts, while considering animals' needs and participation in the design process.